Abba Otieno Ndomo

Atlanta, GA | 404-819-5945 | abba.ndomo@gmail.com | Portfolio: https://andomo3.github.io/portfolioPage/
Work Authorization: F-1 | CPT available now | OPT (May 2027) | STEM OPT eligible

EDUCATION

Georgia Institute of Technology Atlanta, GA
Bachelor of Science in Computer Science, Industrial & Systems Engineering *Aug. 2023 – May 2027*

- **Honors & Awards:** Dean's List, Peer Inspiration Award, 3rd Place @ OKSEF International Engineering Fair
- **Coursework:** Stochastic Processes, Linear & Non-Linear Optimization, Machine Learning, Data Structures & Algorithms, Database Systems
- **GPA:** 3.41

PROFESSIONAL EXPERIENCE

Data Engineer - Altura Jan. 2026 – Present
Python, PySpark, Airflow, pandas, PyArrow, DuckDB, Next.js, Parquet *Atlanta, GA*

- Built production-grade ETL pipeline processing nested Excel pricing matrices across 13 renovation categories into normalized Parquet data lake with 4-dimension tables; reduced estimate generation time from hours to sub-5 seconds for 101-unit property portfolio
- Architected scalable data warehouse schema (star schema design with PySpark-ready Parquet format) supporting 15% faster query performance and seamless horizontal scaling from single-property to multi-property operations across 500+ units
- Engineered Airflow orchestration pipeline (FileSensor → PySpark transformation → DuckDB validation → serving layer) with automated data quality checks, dependency management, and failure recovery—processing 10K+ rows daily with 99% pipeline reliability
- Implemented comprehensive data validation framework with DuckDB catching 95% of schema violations pre-processing; reduced downstream ETL failures from 15% to <1% through type checking, referential integrity validation, and completeness metrics
- Designed modular pricing data model with dynamic tier dimension (Value 0.8x / Standard 1.0x / Premium 1.3x) enabling rapid scenario modeling; supporting 100+ estimate variations daily with sub-200ms query latency through composite indexing

SWE Intern – KPMG Jun. 2024 – Aug. 2024
Python, PostgreSQL, Statistical Analysis, System Design *Mumbai, India*

- Architected fault-tolerant 3-tier data processing pipeline serving 50+ auditors; ingested, transformed, and classified 10,000+ documents monthly with 95% classification precision (validated on held-out test set, $p < 0.001$)
- Optimized end-to-end ETL latency from 45s → 8s (82% improvement) through algorithmic optimization ($O(n^2)$→$O(n+m)$ complexity reduction), parallel batch processing with Python multiprocessing, and database optimization (composite indexes, connection pooling, query restructuring)
- Built production data quality framework implementing 50+ automated validation checks (distributional tests, outlier detection via Z-scores, completeness metrics, referential integrity); reduced data errors from 40% → 5% and prevented 12+ downstream processing failures
- Designed scalable PostgreSQL data model with 3NF normalization across 15 tables; implemented comprehensive testing infrastructure achieving 99% code coverage with CI/CD integration validating data transformations on every deployment

Quantitative Analyst - GTSF Jan. 2025 – May 2025
Python, Bloomberg Terminal API (blpapi), pandas, Statsmodels, NumPy, ARIMA, Monte Carlo *Atlanta, GA*

- Built automated data ingestion pipeline integrating Bloomberg Terminal API; extracted, transformed, and loaded real-time financial data for 120 stocks (2.4x coverage expansion) with sub-minute latency—enabling 30% improvement in portfolio diversification analysis
- Engineered time-series analytics pipeline processing historical price data through ARIMA models and rolling correlation analysis; ran 10,000-iteration Monte Carlo simulations stress-testing portfolio performance under macro shock scenarios, reducing portfolio VaR by 15%

PROJECTS

Founder - PerChance Jan. 2026 – Present
PySpark, Airflow, PyTorch, XGBoost, Django, Postgres, Docker, Redis, React

- Architected end-to-end data pipeline serving 1000+ daily users with 500,000+ daily API calls; automated data ingestion, transformation, feature engineering, model training, and prediction serving via orchestrated Airflow DAGs
- Built scalable PySpark ETL pipeline processing 500,000+ NBA statistical records with custom Holt-Winters time-decay feature engineering; Airflow orchestration of full ETL → feature store → model retraining → serving pipeline reduced ML prep time 90% (from 2 hours to 12 minutes)
- Designed high-throughput 3-tier data architecture (Django API / PostgreSQL database / React frontend) with Redis caching layer and Celery async workers; composite indexes on high-cardinality columns and query optimization sustain sub-200ms p95 API latency under live production traffic
- Implemented comprehensive data observability with Prometheus/Grafana monitoring pipeline health metrics (row counts, null rates, processing latency, error rates); maintained 99.5% pipeline uptime serving 100+ daily users with zero data incidents in 6 months
- Engineered hybrid ML pipeline combining PyTorch deep embeddings with XGBoost gradient boosting; trained on 6 custom time-decay features achieving 70% prediction accuracy across 20 statistical metrics—15% improvement over baseline models

LEADERSHIP

Housing Office Assistant, Lead Conference Assistant Sept.2025 – Present
Georgia Institute of Technology Housing Department *Atlanta, GA*

- Led Delivery & Mail Operations Team; Developed Standard Operating Procedures to coordinate 50+ monthly events serving 1000+ students, achieving 15% improvement in Average Delivery Time

TECHNICAL EXPERTISE

Finance & Quant: Bloomberg Terminal API, Time Series Analysis (ARIMA), Monte Carlo Simulation, VaR, Portfolio Optimization
Languages: Python (OOP, Pandas, NumPy), SQL (CTEs, Window Functions), Java, JavaScript, TypeScript, R, Bash
Data Engineering: Airflow, PostgreSQL, DuckDB, Parquet, PyArrow, ETL/ELT Pipelines, Data Modeling (3NF, Snowflake), Data Warehousing, EERD
Frameworks & Dev: Django, Flask, React, React Native, Next.js, Docker, Redis, Celery, REST APIs, CI/CD, PyTest, Git
Machine Learning: PyTorch, XGBoost, Scikit-Learn, SHAP, Feature Engineering, A/B Testing, Experiment Design, RAG Pipelines, NLP (Transformers)